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Special Collection:

Advances in large-scale hydrological modeling and prediction under global change

Key Points:

- Long Short-Term Memory (LSTM) is applied to estimate runoff at *grid cells*, which is then routed across channel networks with the RAPID model to estimate discharge
- Grid LSTM-RAPID mildly underperforms traditional Basin LSTM but offers vastly greater coverage and outperforms process-based model
- A high-quality daily discharge data set for millions of river reaches worldwide from 1980 to near present is created and shared

Supporting Information:

Supporting Information may be found in the online version of this article.

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Global Daily Discharge Estimation Based on Grid Long Short-Term Memory (LSTM) Model and River Routing

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Abstract To expand the spatial coverage of the conventional Basin Long Short-Term Memory (LSTM) model for river discharge estimation beyond pre-selected individual locations, we developed a discharge modeling scheme, Grid LSTM-RAPID, to estimate discharge for every river reach worldwide. Grid LSTM-RAPID applies LSTM runoff estimation to the grids (0.25°), small rectangular hydrological response units (HRUs) rather than basins (irregularly shaped HRUs of any size), and then routes the grid runoff over all reaches on a global river network using the RAPID routing model. It largely maintains the strong performance of Basin LSTM over gauged basins and achieves a median Kling-Gupta Efficiency (KGE) of 0.653 for small basins out-of-sample both temporally and spatially (0.688 for out-of-sample temporally), and a median KGE of 0.592 for other basins with larger areas and less data quality. Compared to Basin LSTM, Grid LSTM-RAPID loses about 0.03 in median KGE for basins out-of-sample in both time and space in exchange for global all-reach coverage without heavy cost. Despite this tradeoff, it significantly outperforms a well-calibrated process-based benchmark model. Using the new scheme, we create an improved global reach-level daily discharge data set from 1980 to near present named GRADES-hydroDL, which is openly shared at <https://www.reachhydro.org/home/records/grades-hydrodl>.

1. Introduction

River discharge plays a crucial role in the global water cycle (Wada et al., 2017). Accurate global river discharge estimation is of vital importance across various fields like water resources (Liu et al., 2020; Oki & Kanae, 2006), climate change (Gerten et al., 2008; Milly & Dunne, 2020; Trabucco et al., 2008), natural hazards (Coughlan de Perez et al., 2016; Yang et al., 2021), biodiversity (Ficke et al., 2007; Vörösmarty et al., 2010) and energy production (Chen et al., 2016; Xu et al., 2023). Normally, gauging stations are deemed the most reliable data source for river discharge (Fekete et al., 2002; Zaitchik et al., 2010). However, a significant proportion of the world's rivers remain ungauged due to a combination of technical, economic, and political constraints (Hannah et al., 2011; Riggs et al., 2023). Recent advancements in remote sensing (RS) have opened new avenues for global river discharge monitoring, even in ungauged basins (Biancamaria et al., 2016; Bjerklie et al., 2018; Yang, Lin, et al., 2019). Nonetheless, the temporal coverage of satellite observations is largely limited to the recent two decades and RS-based discharge estimations hinge upon prior knowledge of river discharge to reduce uncertainties and improve accuracy (Durand et al., 2023; Tuozzolo et al., 2019).

As a result, considerable efforts have been made by the community to estimate river discharge based on various rainfall-runoff models. The existing rainfall-runoff modeling approaches can be roughly categorized into data-driven and process-based approaches. Historically, process-based models are the main choice for discharge estimation. A pathway pioneered in large-scale hydrology has been to utilize advanced process-based models, together with the optimal combination of in situ and satellite observations, as well as reanalysis, to reconstruct river discharge globally (Alfieri et al., 2020; Harrigan et al., 2020; Hersbach et al., 2020). For example, the European Commission's Copernicus Emergency Management Service Global Flood Awareness System (Glo-FAS) employs a coupled land surface model (the Hydrology Tiled ECMWF Scheme for Surface Exchanges over Land - HTESSEL) and flow routing model (LISFLOOD) to generate long-term river discharge at daily, 0.05°

resolution (Alfieri et al., 2020; Harrigan et al., 2020). Using the Variable Infiltration Capacity (VIC) model (Liang et al., 1994, 1996) that is well calibrated and bias-corrected and the Routing Application for Parallel computation of Discharge (RAPID) river routing model (David et al., 2011), Lin et al. (2019) produced the first reach-level naturalized daily river discharge, the Global Reach-Level A Prior Discharge Estimates for SWOT (GRADES), over 2.94 million river reaches globally for 1979–2014. Building upon the GRADES legacy, Yang et al. (2021) pushed the temporal resolution down to 3-hourly, creating the Global Reach-Level Flood Reanalysis (GRFR) product for flood analysis. Building on the foundations of GRFR and GloFAS, and assimilating river discharge observations derived from Landsat images, Feng and Gleason (2024) developed the Global River Discharge Reanalysis (GRDR) data set.

Recently, the availability of extensive data sets and advancements in computing technologies have facilitated the development of data-driven techniques, predominantly based on machine learning (ML). These ML models directly learn intricate non-linear response patterns from massive amounts of data, without requiring explicit knowledge of the underlying physical processes and strong structural assumptions (LeCun et al., 2015; Prasad et al., 2017; Schmidhuber, 2015; Shen, 2018; Shen et al., 2023). These advances have also drawn the attention of the hydrological community, inspiring new efforts to apply ML models to rainfall-runoff modeling. Among many efforts, a popular model is the Long Short-Term Memory (LSTM) neural network, a specifically designed version of recurrent neural network for long-term sequential data sets (Greff et al., 2016; Hochreiter & Schmidhuber, 1997). With long-term memory, LSTM excels in capturing both periodic and chaotic behaviors within time-series data, as well as learning their long-range dependencies with higher accuracy (Fang et al., 2017; Hu et al., 2019; Mouatadid et al., 2019). This makes LSTM particularly suitable for hydrologic modeling and sparks a proliferation of research on LSTM-based rainfall-runoff modeling. Multiple researchers have demonstrated LSTM's seemingly incomparable performance in simulating runoff (Feng et al., 2020, 2021; Frame et al., 2022; Gauch et al., 2021; Konapala et al., 2020; Kratzert et al., 2021; Kratzert, Klotz, Shalev, et al., 2019; Lees et al., 2021; Nearing et al., 2021; Reichstein et al., 2019; Sun et al., 2021). Most applications focus on data-rich regions and some focus on the globe (Feng et al., 2024; Nearing et al., 2024; Rasiya Koya & Roy, 2024; Tang et al., 2023).

The approach employed in these studies to train and apply LSTM over pre-selected basins, treats each basin (of any shape and size) as a hydrological response unit (HRU) and utilizes basin-average inputs to estimate the discharge only at the basin outlets. We refer to this approach as Basin LSTM. While Basin LSTM has shown promise, the approach faces several challenges. First, since Basin LSTM only simulates discharge at the basin outlet, it is cumbersome to apply the model to a large number of river reaches (millions globally). Every new location, no matter how close to existing ones, would require re-aggregating all the data over a different polygon. Second, this approach relies on basin-averaged inputs (e.g., precipitation and temperature) (Lees et al., 2022). This basin averaging procedure eliminates spatial variability within, causing some concerns over very large basins, such as Amazon, Mississippi, where factors like storm locations may matter (Hunt et al., 2022; Le et al., 2019; Wang & Karimi, 2022). Third, consistency across HRUs, such as the up/downstream mass balance and their temporal coherence, can be better constrained if we have an explicit river routing process (Wang et al., 2024) instead of modeling parent/children basins separately. Note that mass balance is not the focus of this study, nor is it a critical matter. Also, recent research has begun to integrate physics into ML frameworks to address this limitation (Feng et al., 2024; Wang et al., 2024). Overall, we still need to work on making LSTM capable of global discharge estimation for millions of reaches.

To meet the challenge, we propose to train LSTM over small basins, apply it for runoff estimation at grid cells, and route the gridded runoff across the channel network with the RAPID routing model—a new scheme referred to as Grid LSTM-RAPID. The concept of Grid LSTM builds upon the above discussed Basin LSTM, but differs in the type of HRUs to which the LSTM is applied: while Basin LSTM is applied to irregularly shaped basins (Frame et al., 2025), Grid LSTM applies the model to rectangular grids. This grid-based structure enables the direct use of global gridded inputs to force the trained LSTM model and generate spatially distributed runoff. By integrating grid-based runoff estimation with network-based flow routing, Grid LSTM-RAPID offers a framework for global discharge estimation. Clearly, the LSTM training and application will be performed on mismatched parameters—small basin per-area discharge versus grid runoff, and the adding of routing modeling will introduce some errors. We will study the pitfall/price of doing so. In later sections, we will cover the methodology, data, and evaluation experiments, discuss the performance gains/losses against Basin LSTM, as well as a process-based model, and introduce a global river discharge data set from the new scheme.

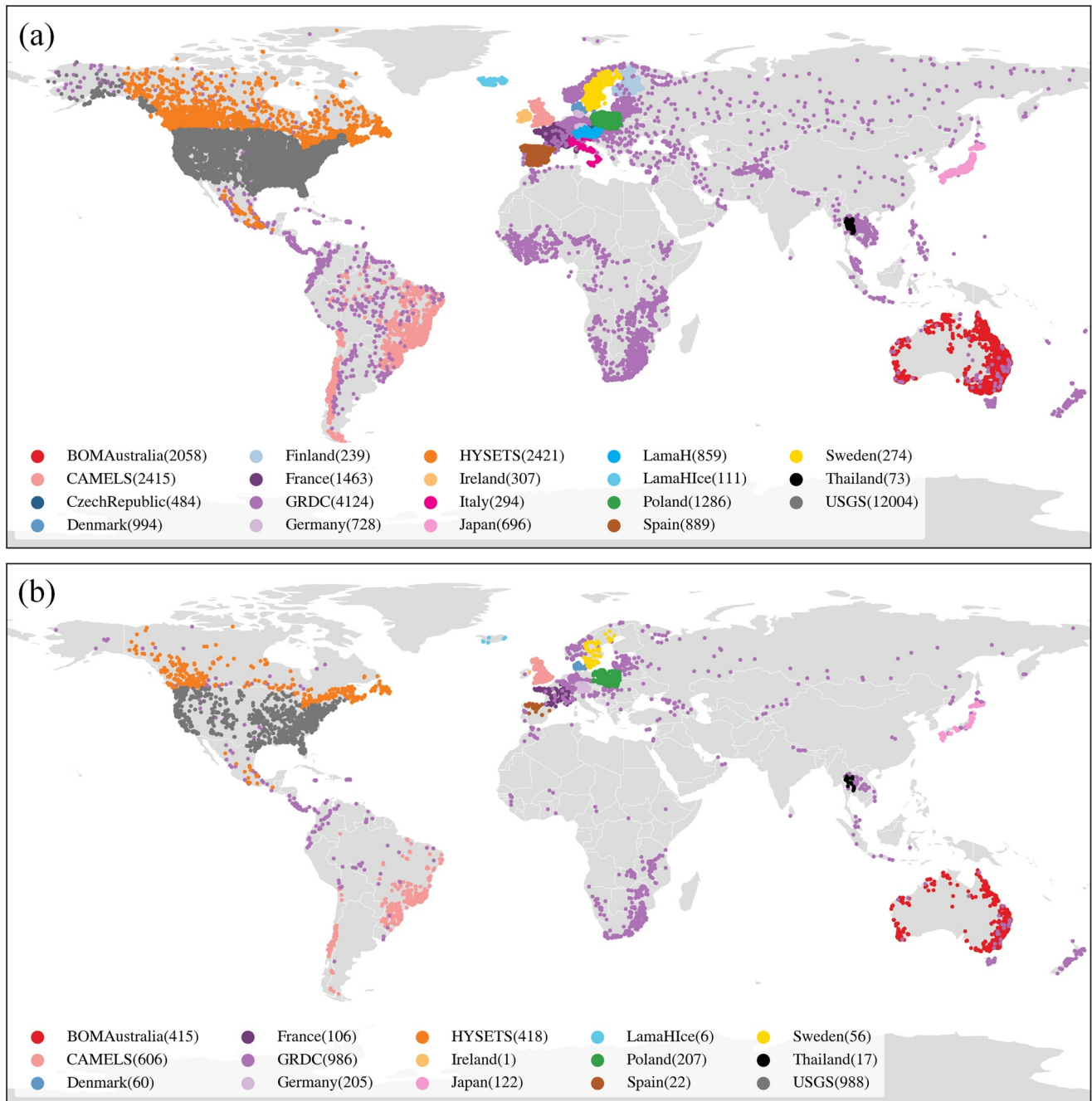


Figure 1. Maps of (a) all basins and (b) selected training basins. The legend shows the data sources (see details in data availability) and the number of gauges.

2. Methodology

2.1. Data

The study requires dynamic meteorological forcings, static basin attributes, river discharge observations, as well as a hydrography data set for global rivers.

Dynamic meteorological forcings are the driving inputs for hydrologic models. We use the precipitation from Multi-Source Weighted-Ensemble Prediction (MSWEP) version 2.80 (Beck et al., 2018), and other meteorological fields from ERA5 (Hersbach et al., 2018), as well as monthly leaf area index (LAI) climatology (no-interannual change) from PROBA-V (Fuster et al., 2020) (Table S1 in Supporting Information S1). MSWEP

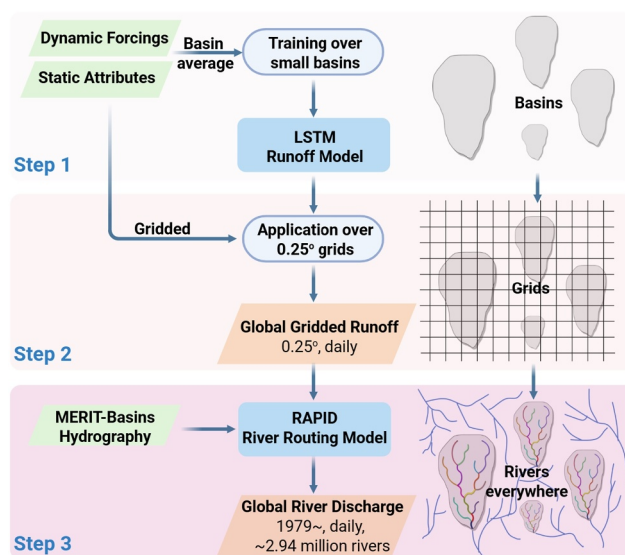


Figure 2. Grid LSTM-RAPID daily discharge modeling framework. The basins, grids, and river networks depicted are simplified illustrations and do not represent their actual size, shape, or quantity.

combines in situ gauges, satellite products, and reanalysis products, and makes distributional bias corrections, as well as corrections of systematic terrestrial biases using river discharge observations. ERA5 is the latest reanalysis from the European Center for Medium-Range Weather Forecasts (ECMWF). Monthly LAI climatology is used in this study to account for vegetation changes, which have been shown to benefit discharge estimation (Tounsi et al., 2023). However, this climatology simplifies the representation of vegetation dynamics and may overlook the interannual variability, potentially introducing uncertainty into discharge simulation.

To inform LSTM about basin rainfall-runoff behaviors, we calculate the top 10 sensitive basin attributes according to Kratzert, Klotz, Shalev, et al. (2019), including climate, topography, and soil attributes (Table S1 in Supporting Information S1). These attributes are static and appended to the forcing data as input for LSTM.

Globally, we compile daily discharge records for 31,719 river gauges from multiple public sources, including the United States Geological Survey (USGS) National Water Information System (NWIS), Global Runoff Data Center (GRDC), a series of Catchment Attributes and MEteorology for Large-sample Studies (CAMELS) data sets, etc. North America and Europe have much higher gauge densities (Figure 1a). Discharge records in Africa and most regions of Asia are only from GRDC gauges, most of which are in large basins or provide only monthly data or limited data before 1995. This poses a challenge to model training and evaluation in these regions.

The vector hydrography data set MERIT Basins (Lin et al., 2019) is used for routing. MERIT Basins is derived from the 3 arc-second (~ 90 m) MERIT-Hydro product (Yamazaki et al., 2019), using 25 km^2 accumulation area as the channelization threshold. It contains ~ 2.94 million river reaches and corresponding catchments with derived properties (e.g., slope, length, connectivity).

2.2. Grid LSTM-RAPID (and Basin LSTM)

The Grid LSTM-RAPID scheme consists of three steps (Figure 2):

Step 1. LSTM training over small basins: Train a single LSTM model over selected training basins, a widely used routine in current LSTM-based discharge estimate studies (Feng et al., 2020; Nearing et al., 2024). Additional LSTM details are Text S1 in Supporting Information S1.

This step uses basin-averaged inputs, thus also trains a traditional Basin LSTM. We also directly apply this trained Basin LSTM to estimate discharge at the basin outlets for comparison purposes. The target variable is a dimensionless discharge value (discharge normalized by basin area and mean daily precipitation). More details of LSTM training are in Sections 2.3 and 2.4.

Step 2. LSTM application over 0.25° grids: Apply the trained LSTM obtained in Step 1 over global 0.25° grids, using the gridded inputs, to generate global gridded runoff. In other words, we apply the rainfall-runoff relationship learned by the LSTM model in the training basins to grid cells. The areas of global 0.25° grid cells range from 85.8 to 772.8 km², which falls within the range of areas of training basins (Figure S2a in Supporting Information S1). This overlap supports the feasibility of the application of the learned model.

Step 3. RAPID routing: Implement the RAPID model to calculate the discharge for all reaches globally on the MERIT-Basins river network based on the gridded runoff from Step 2. RAPID uses a vector-matrix version of the Muskingum method and is well parallelized for large-scale applications. RAPID setup details can be found in Yang et al. (2021).

2.3. Training Basin Selection

We perform a screening to identify suitable training basins globally, using several criteria including boundary accuracy, basin area, data length, low human influences and visual inspection (Text S2 in Supporting Information S1). The selection procedure finally produces a list of 4215 basins for LSTM training (Figure 1b).

2.4. LSTM Implementation

A single global LSTM model is trained using training period data from all training basins so that the network can learn a more general understanding of the rainfall-runoff process. The LSTM network is trained on sequences of 365 days of six meteorological features and 10 static basin attributes (Table S1 in Supporting Information S1) to simulate the discharge at daily scale. The loss function is the Root-Mean-Squared Error (RMSE), calculated on the transformed daily discharge (see more details in data preprocessing in Text S3 in Supporting Information S1), aimed at improving low flow representation. Hyperparameter combinations (Table S2 in Supporting Information S1) from Feng et al. (2020) are utilized, and through a simple validation process, it is determined that these hyperparameters remain optimal for discharge estimation in this study. A fast and flexible LSTM code from the open source hydroDL repository is implemented. It takes about 8 hr on an NVIDIA A100 Graphical Processing Unit to reach convergence (300 epochs) for 4215 basins with 20 years of training data.

Considering the stochasticity in neural network training, we perform an ensemble of four randomly seeded LSTM trainings and the mean of all four LSTM predictions is used.

2.5. VIC-RAPID: The Process-Based Streamflow Modeling Scheme as a Benchmark

For benchmarking/comparison purposes, we also implement a process-based global streamflow modeling scheme VIC-RAPID, where the VIC model is used for gridded runoff calculation and RAPID for routing. The meteorological forcing remains the same (MSWEP V2.80 and ERA5). To reduce model biases, grid-level parameter calibration (Yang, Pan, et al., 2019) and bias correction are performed against ML-derived, global runoff characteristic maps from the Global Streamflow Characteristics Data set (GSCD; Beck et al., 2015). More details can be found in Yang et al. (2021). To our knowledge, this is one of the best global simulations from process-based models (Tu et al., 2024).

2.6. Evaluation Experiments

We evaluate all three discharge estimation schemes implemented in the study: (a) Basin LSTM, (b) Grid LSTM-RAPID, and (c) VIC-RAPID. Basin LSTM sets an upper bound of LSTM performance since it's trained and applied consistently at the same HRUs. Comparing Grid LSTM-RAPID with Basin LSTM helps us understand the skill tradeoff as well as verify the overall quality of data and LSTM setup relative to other studies in the literature.

To study the behavior of the different methods in ways comparable to various previous studies of reference (Feng et al., 2024; Nearing et al., 2024; Rasiya Koya & Roy, 2024), we split the data for training/testing across two dimensions:

1. Temporal split: The whole data is split into two periods: 1980–1999 and 2000–2020. 1980–1999 for training, 1980–1999 and 2000–2020 for testing separately.
2. Spatial split:

Table 1
Evaluation Experiments

Estimation schemes to test	Spatial	Temporal	
		In-sample (1980–1999)	Out-of-sample (2000–2020)
Basin LSTM	In-sample	Training	Temporal test
Grid LSTM-RAPID			
VIC-RAPID			
Basin LSTM	10-fold cross-validation	Spatial test	Spatial-temporal test
Grid LSTM-RAPID			
Grid LSTM-RAPID	Extended out-of-sample	Extended spatial test	Extended spatial-temporal test
VIC-RAPID			

- (i) 10-fold cross-validation split: randomly split the 4215 well-selected high-data-quality small basins into 10 groups, select 9 groups for training and 1 for testing, and rotate the selection 10 times (see groups B01–B10 in Figure S1 in Supporting Information S1).
- (ii) Extended-sample bulk split: here we extend the testing from 4215 small basins to all gauging locations regardless of basin size, that is, 4215 small basins for training and the rest 27,504 basins of any size and data quality for testing (group A in Figure S1 in Supporting Information S1). Obviously, this test is much more challenging than the 10-fold cross-validation within small basins due to factors like the extremely small training-testing sample ratios, large scaling differences, and low data quality. This test is needed to understand the added value of Grid LSTM-RAPID (and any routing-based methods) though it is not conducted for Basin LSTM due to difficulties in scaling it up computationally.

Depending on whether we split the data temporally (in-sample/out-of-sample) and whether/how we split the data spatially (in-sample/10-fold cross-validation/extended out-of-sample), we conduct $2 \times 3 = 6$ evaluation experiments on different discharge estimation schemes (Table 1). Each experiment is given a name according to the nature of its testing sample: training, temporal test, spatial test, spatial-temporal test, extended spatial test, and extended spatial-temporal test.

Note that VIC is tuned/bias-corrected globally over a longer period (Yang et al., 2021) and the experiment names for VIC-RAPID are only used for comparison. This makes its comparison against Grid LSTM-RAPID somewhat unfair for the latter.

Since Grid LSTM-RAPID and VIC-RAPID provide discharge for river reaches, we first match gauges with the corresponding river reaches (Text S4 in Supporting Information S1). For all evaluation experiments, we retain results only for basins with more than 2 years of valid data during the respective evaluation period. The basin numbers used in various simulations and evaluation experiments are shown in Table S3 in Supporting Information S1.

Metrics adopted to evaluate model performance include the modified Kling-Gupta Efficiency (KGE, Kling et al., 2012) and its three component metrics: correlation coefficient (CC, for temporal coherence), relative variability (RV, for bias in variability), and relative bias (RB, for bias in magnitude). We also calculate the percent bias of the top 2% peak flow range (FHV) and the percent bias of the bottom 30% low flow range (FLV) (Yilmaz et al., 2008), to highlight the performance of the model for peak flows and baseflow. For Basin LSTM and Grid LSTM-RAPID, all metrics are calculated based on the ensemble-mean discharge.

3. Results

3.1. Basin LSTM

Basin LSTM helps us understand the upper bound of LSTM performance and provides a sanity check for obvious mistakes in the data and model setup. In general, the overall performance will naturally degrade along with the inclusion of out-of-sample data. The median KGEs of Basin LSTM predictions are 0.776, 0.755, 0.673 and 0.674, in training, temporal test, spatial test, and spatial-temporal test (Figure 3). With more out-of-sample data in the

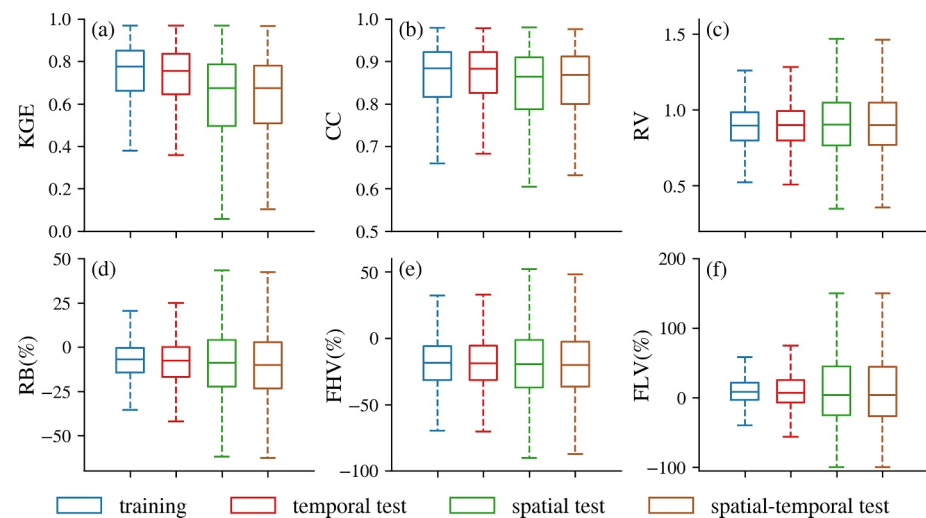


Figure 3. Performance of the four evaluation experiments for Basin LSTM: (a) KGE, (b) CC, (c) RV, (d) RB, (e) FHV, and (f) FLV.

test, CCs tend to decrease while median RBs/RVs remain stable though with more spread. We notice two kinds of median KGE drops: (a) ~ 0.02 between temporal test and training and (b) ~ 0.1 between spatial test and training. This suggests that it's more challenging for LSTM to generalize its efficacy to spatially unseen data than temporally unseen data (Feng et al., 2021; Song et al., 2024). No clear relationship is observed between model performance and basin area, as evidenced by the scatter plot of KGE versus basin area in the spatial-temporal test (Figure S2b in Supporting Information S1). Similar patterns are also seen in the temporal and spatial tests. These Basin LSTM performance numbers are comparable to findings in other studies. For example, Rasiya Koya and Roy (2024) achieved a median KGE of 0.679 for temporal test over 2610 basins. Tang et al. (2023) reported a median Nash–Sutcliffe Efficiency (NSE) of 0.59 for temporal test over 1897 basins with areas larger than 9,000 km² (our median NSE for temporal test is 0.74). Feng et al. (2024) obtained a median KGE of 0.74 for temporal test, 0.67 for spatial test over 3000+ basins. Nearing et al. (2024) achieved a median KGE of 0.643 for spatial-temporal test (calculated by us from their data).

3.2. Tradeoff Between Basin LSTM and Grid LSTM-RAPID

Grid LSTM-RAPID behaves quite similarly to Basin LSTM in the relative changes of CC/RV/RB between experiments (Figure S3 in Supporting Information S1 and Table 2). The median KGE drops by 0.01 from training to temporal test and 0.04 to spatial test (0.02 and 0.1 in case of Basin LSTM). The less drop in spatial test suggests that routing may help the model to generalize its efficacy spatially.

The median KGE values for training, temporal, spatial, and spatial-temporal tests for both Basin LSTM and Grid LSTM-RAPID are shown in Table 2. As fewer basins are evaluated in Grid LSTM-RAPID (due to the matching between river reaches and gauges, see Table S3 in Supporting Information S1), the Basin LSTM metrics are recalculated for the same basins for a fair comparison (numbers slightly different from Section 3.1). The median KGE of Grid LSTM-RAPID is lower than Basin LSTM, by about 0.07–0.08 in the training and temporal test, and about 0.03 in spatial and spatial-temporal tests. This performance loss is the price of applying the LSTM learned from irregular-shaped basins to rectangular grids as well as adding the routing. In exchange, Grid LSTM-RAPID is able to model every river reach without heavy computations. The performance penalty is expected but the level in temporal test (~ 0.07) is still quite pronounced (although the level in spatial test, ~ 0.03 , is rather small). Basin LSTM works entirely at basin scale thus bypasses the uncertainties in the sub-basin scale processes. Grid LSTM-RAPID suffers from these sub-basin scale errors, for example, routing errors, which apparently can be significant. We think this situation can be improved, for example, a more trainable routing model than RAPID is needed that is better tuned against discharge observations at basin outlets. Emerging efforts like differentiable routing (Bindas et al., 2024) are tackling this very issue.

Table 2
Median KGEs of All Evaluation Experiments

	Basin LSTM	Grid LSTM-RAPID	VIC-RAPID
Training (2940)	0.783	0.699	0.481
Temporal test (2642)	0.761	0.688	0.491
Spatial test (2940)	0.689	0.660	–
Spatial-temporal test (2642)	0.687	0.653	–
Extended spatial test (3852)	–	0.596	0.440
Extended spatial-temporal test (4510)	–	0.592	0.443

Note. The number of basins tested is noted in parentheses.

Figure 4 compares the histograms of skill metrics for the most challenging spatial-temporal test from the two models. Grid LSTM-RAPID shows significant degradation over Basin LSTM, as indicated by p-values below 0.05 from the Kolmogorov-Smirnov test (K-S test; Eghbali, 1979; Smirnov, 1948). Besides the 0.03 median KGE loss (from 0.687 to 0.653, Table 2), the fraction of basins with KGE > 0.7 drops from 47% to 40%. Most of the degradation comes from lower CC and underestimated variability (Figure 4). Variability is already underestimated in Basin LSTM and becomes worse in Grid LSTM-RAPID, likely due to the difficulty for LSTM model to pick up a faster runoff dynamic. Grid LSTM-RAPID is slightly lower-biased though.

3.3. Global Grid LSTM-RAPID Performance

Extended spatial and spatial-temporal tests provide a global look into model applicability. Skill metrics are expected to further drop as dramatically more and fully independent basins of larger size and less data quality are included. The results in Table 2 show a median KGE drop of about 0.06 and wider spreads as the spatial/spatial-temporal tests are extended globally. This drop is a result of lower CC and more spreads in RB and RV (Figure S3

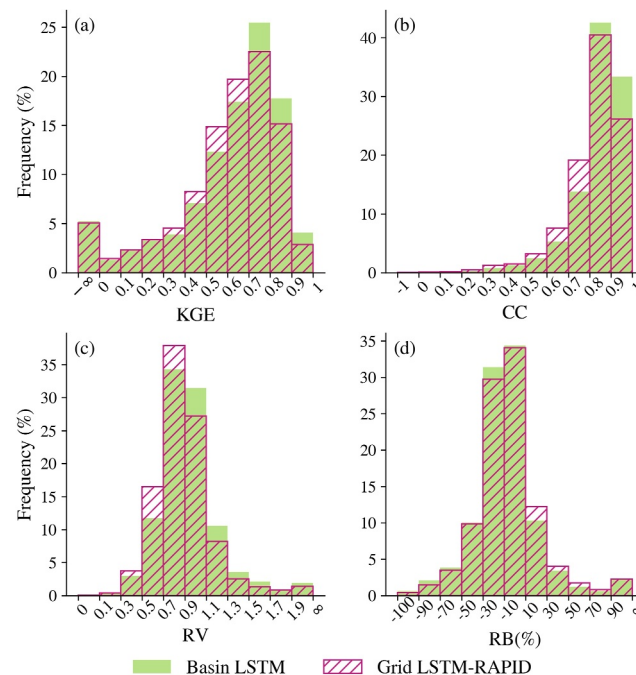


Figure 4. Histograms of (a) KGE, (b) CC, (c) RV, and (d) RB for the spatial-temporal test from Basin LSTM (green) and Grid LSTM-RAPID (red). The differences are statistically significant with p-values less than 0.05 based on the K-S test.

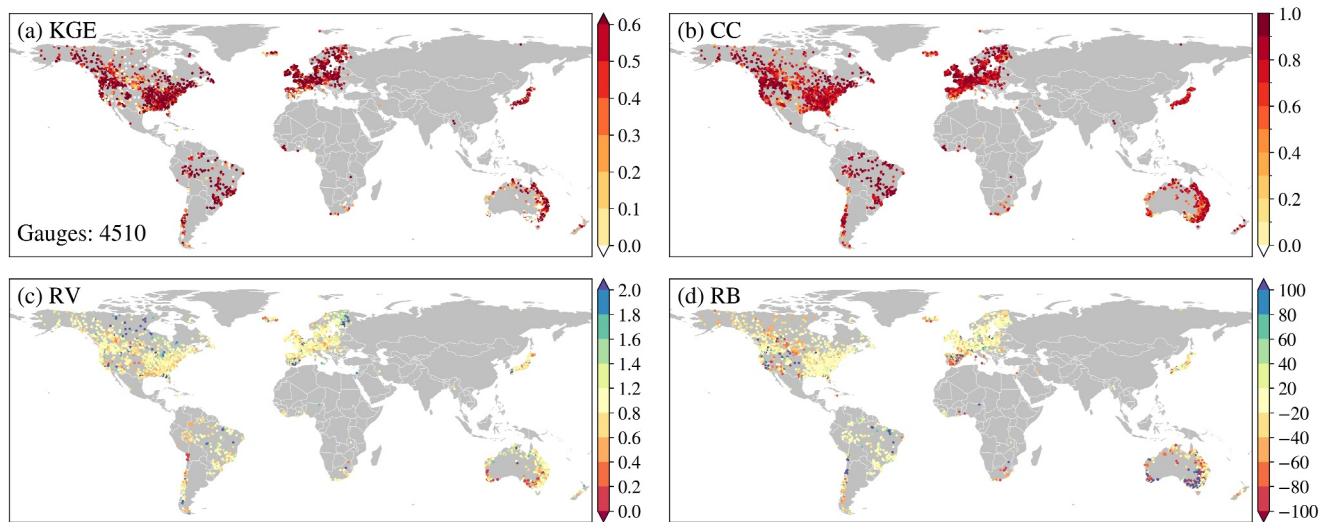


Figure 5. Skill metrics of extended spatial-temporal test from Grid LSTM-RAPID: (a) KGE, (b) CC, (c) RV, and (d) RB.

in Supporting Information S1). Nonetheless, the median KGE remains at 0.596 and 0.592 for the two tests, which is still competitive for global river discharge simulations.

More details about the most challenging experiment of globally extended spatial-temporal test on Grid LSTM-RAPID are shown in Figure 5. The discharge dynamics are well simulated over most of the world with $CC > 0.6$ over 88.7% of the locations. Arid areas like the central CONUS, South Africa, and parts of Australia show lower CCs. Discharge variability is well captured except for some tendency to underestimate in Australia and overestimate in northern Canada and Finland. The total flow volume is also well reproduced with about 52.1% (74.6%) gauges having RB within $\pm 20\%$ ($\pm 40\%$). Overall, about 62.3% of locations show $KGE > 0.5$. Similar to previous model-based studies (Alfieri et al., 2020; Lin et al., 2019; Tu et al., 2024; Yang et al., 2021), Grid LSTM-RAPID also struggles with many arid regions.

3.4. Comparison Between Grid LSTM-RAPID and VIC-RAPID

Grid LSTM-RAPID exhibits overwhelming advantages over the physically based VIC-RAPID for all four overlapping experiments (training, temporal test, extended spatial test and extended spatial-temporal test) (Table 2), with the median KGE difference about 0.15–0.22. Note that VIC is always globally trained over a longer period and the new scheme earns its advantages under an unfavorable condition. More detailed skill comparisons in the most challenging extended spatial-temporal test can be found in the histograms in Figure 6, where significant improvements in KGE can be mostly attributed to better CC (temporal dynamics), and less to lower variability bias and runoff bias (since VIC is already bias-corrected).

Maps of performance differences in the same test are shown in Figure 7. The locations of Grid LSTM-RAPID advantages are shown as red dots in the left column and locations of VIC-RAPID advantages as blue dots on the right. Counts of locations of significant advantages are labeled at the corner of each map, with locations of small differences (± 0.1 for KGE, CC and $|RV - 1|$, $\pm 10\%$ for $|RB|$) not counted. On the advantaged location counts, Grid LSTM-RAPID excels strongly in CC (2584 vs. 165), less in RV (1547 vs. 1119), and least in RB (1337 vs. 1137). Most improvement in CC is located in the Rocky Mountains, southern Australia and eastern Europe, and RV improvement is largely around the central CONUS and Australia. Beyond that, no strong correlations with topography, climate, and gauge basin area are observed. Despite the clear advantage of a data-driven runoff model, the fact that VIC-RAPID performs better in certain basins (Figure 7b) still underlines the value of physical constraints and process knowledge in a process-based model.

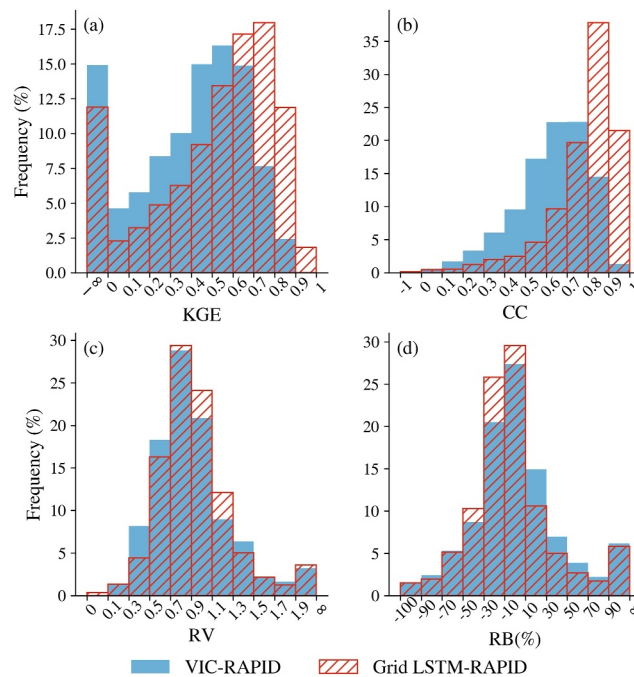


Figure 6. Histograms of (a) KGE, (b) CC, (c) RV, and (d) RB for the extended spatial-temporal test from Grid LSTM-RAPID (red) and VIC-RAPID (blue). The differences are statistically significant with p -values less than 0.05 based on the K-S test.

4. Global Daily Reach-Level River Discharge Data Set Based on Grid LSTM-RAPID (GRADES-hydroDL)

Based on Grid LSTM-RAPID, we develop an improved global reach-level daily database of discharge records from 1980 to near present (2025 so far). The setups are the same as in the extended sample experiment except that all data in 1980–2020 are used in training to maximize skill. This updated data set serves as a new iteration of the previously developed GRADES (Lin et al., 2019), and is referred to as GRADES-hydroDL. Here, hydroDL is a widely applied open-source repository (Aboelyazeed et al., 2023; Chaemchuen et al., 2023; Fang & Shen, 2020; Feng et al., 2020, 2021; Liu et al., 2023; Rahmani et al., 2021; Saha et al., 2023; Shen et al., 2023; Zhi et al., 2023), which consists of a set of consistent hydrologic deep learning implementations using a specific library. The hydroDL library contains models for several versions of LSTM models for hydrologic and water quality variables, multiscale models, physics-informed differentiable models, and in the future Transformer models.

GRADES-hydroDL reproduces the global discharge very well (Figure 8). It shows overwhelming advantages over the GRDR reanalysis data set (Feng & Gleason, 2024), which already assimilates the remotely sensed river discharge observations (Figure S4 in Supporting Information S1). Figure 9a maps its mean daily river discharge from 1980 to 2020 for each river reach with a stream order larger than 5 and discharge value larger than $1 \text{ m}^3/\text{s}$. The pattern is similar to previous results (Harrigan et al., 2020; Lin et al., 2019). Figure 9b shows the inter-annual variability of global rivers, quantified by the coefficient of variation (CV) of annual discharge. Most rivers have $CV < 0.4$, while some regions see substantial inter-annual variability. From the Köppen-Geiger climate classification map (Beck et al., 2018), these high variability rivers are mostly in arid regions and driven by precipitation variability (Fielding et al., 2018; McMahon et al., 1987).

A quick analysis of flood seasonality (Berghuijs et al., 2019; Blöschl et al., 2017; Collins, 2019; Dickinson et al., 2019; Hall & Blöschl, 2018; Villarini, 2016; Ye et al., 2017) is derived from GRADES-hydroDL. Based on the annual maximum flows, two seasonality metrics were calculated: (a) average timing of river flood $\bar{D}$, quantifying the time of the year in which flood events tend to occur and (b) concentration of floods R , quantifying how strong the seasonality is. Higher values denote a more concentrated flood season while lower values signify spread-out seasonal distributions (see Text S5 in Supporting Information S1). The mean seasonality of flood

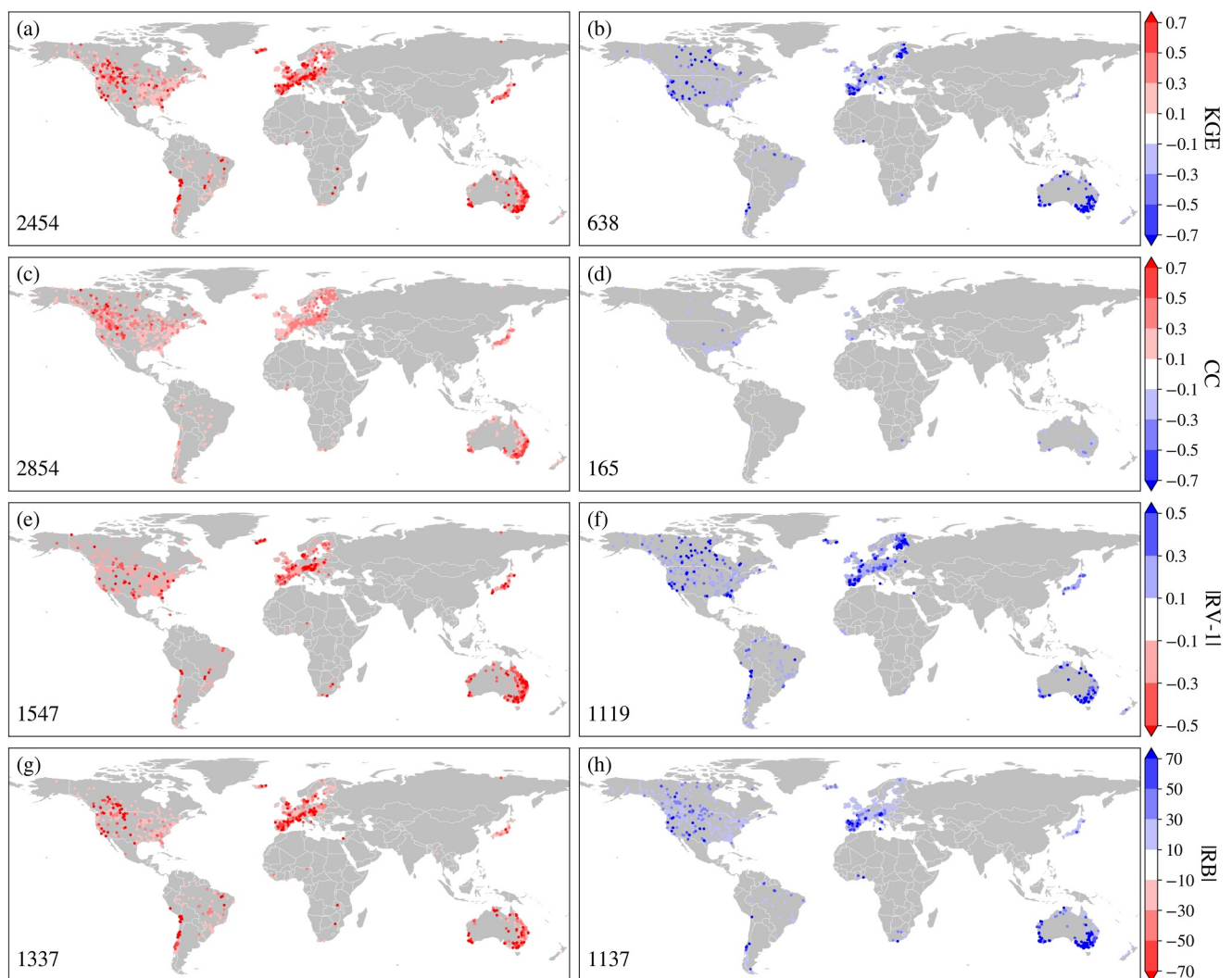


Figure 7. Skill differences for extended spatial-temporal test between Grid LSTM-RAPID and VIC-RAPID (differences in KGE, CC, IRV-II, and IRBI). Red dots indicate that Grid LSTM-RAPID performs better (left column) and blue dots indicate that VIC-RAPID performs better (right column). Counts of locations of significant advantages are labeled at the corner of each map (dots in red/blue color ranges but not in white).

exhibits distinct regional features (Figure 9c). In low-latitude tropical regions, annual floods predominantly occur during the wet summer season of July–September in the Northern Hemisphere and December–February in the Southern Hemisphere. High-latitudes and high-altitudes (e.g., Rockies, Alps) primarily experience late spring and summer floods because of snow storage and melt (Hall & Blöschl, 2018; Parajka et al., 2009; Villarini, 2016; Ye et al., 2017). The seasonal concentration of flood R is shown in Figure 9d. Globally, the seasonality is strong with $R > 0.9$ in most areas. Relatively weaker seasonality is found mainly in the southeastern CONUS, the Mediterranean, southern Brazil, and equatorial regions, with $R < 0.5$.

GRADES-hydroDL can be valuable for investigating water-related issues like flood assessment, water resources management, and ecological/environmental studies. Compared with process-based simulation products (GRADES, GRFR) as well as the reanalysis data set GRDR, it can provide better prior information in support of scientific applications requiring spatiotemporal continuous discharge estimates. In future studies, the updated GRADES-hydroDL shall supersede its predecessor GRADES. We highly recommend using this new data in relevant applications.

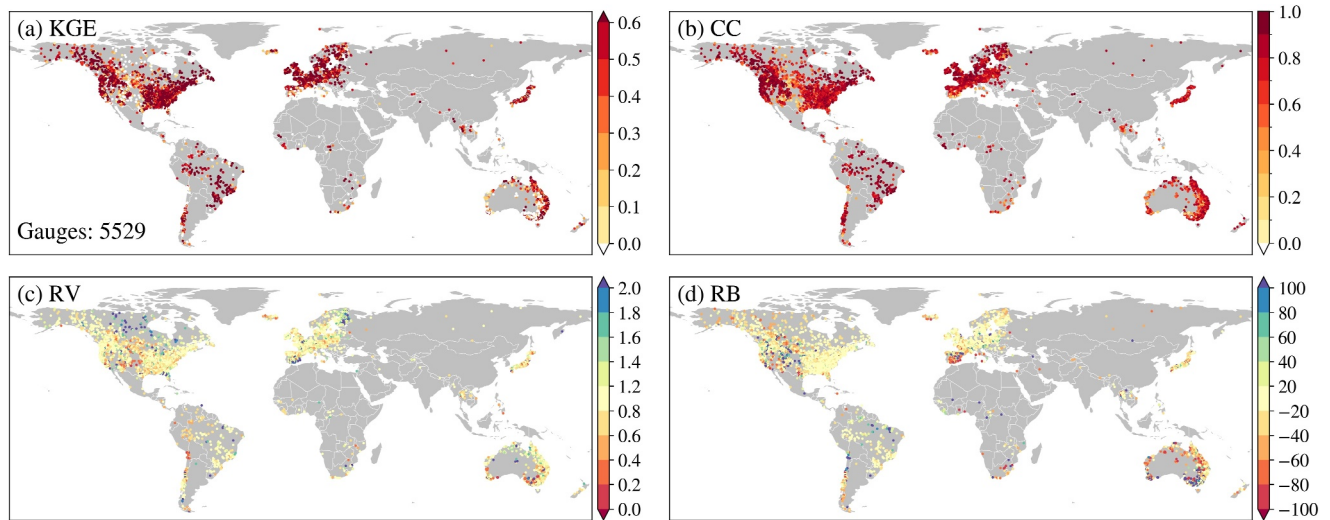


Figure 8. Skill metrics over non-training basins for GRADES-hydroDL during the period of 1980–2020: (a) KGE, (b) CC, (c) RV, and (d) RB.

5. Conclusions

Building on top of Basin LSTM, Grid LSTM-RAPID tries to apply LSTM-based runoff estimation to rectangular HRUs (0.25° grids here), so that river discharge can be estimated everywhere with the help of a routing model. Through the data building, model development/training, and skill verification/analysis effort, we try to study the effectiveness, tradeoffs, and drawbacks of this discharge estimation scheme as well as to create a high-quality long-term global discharge data set for potential uses by the water research community. Results here show that:

1. Our global LSTM implementation and supporting data work reasonably well with basin-level skills and are comparable to those reported from existing studies.
2. Experiments on Grid LSTM-RAPID, as well as Basin LSTM, show how these estimation schemes perform and how well their skills generalize from seen to unseen data in time and space.
3. Skill analysis across estimation schemes reveals that Grid LSTM-RAPID sustains a median KGE skill loss around 0.03–0.08 in various tests against Basin LSTM in exchange for the global all-river-reach coverage.

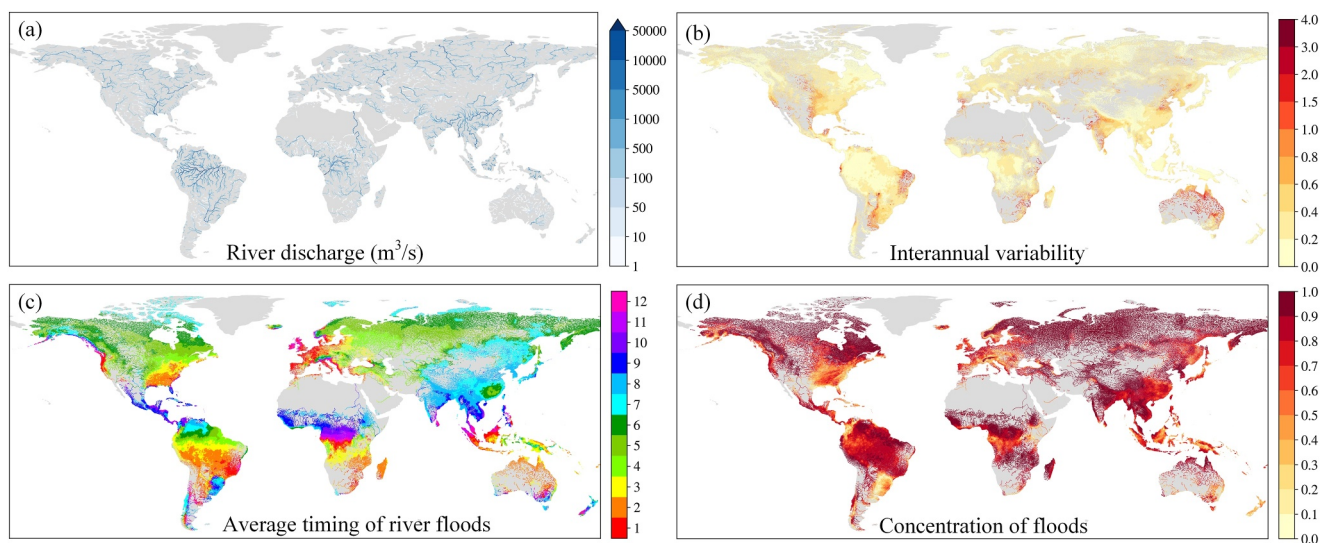


Figure 9. Characteristics for GRADES-hydroDL. (a) Mean daily river discharge during 1980–2020 for river reaches of stream order ≥ 5 and discharge value $\geq 1 \text{ m}^3 \text{ s}^{-1}$, (b) Interannual variability (CV) for river reaches of discharge value $\geq 1 \text{ m}^3 \text{ s}^{-1}$, (c) Average timing of river floods ($\bar{D}$) for river reaches of discharge value $\geq 1 \text{ m}^3 \text{ s}^{-1}$, (d) Concentration of floods within a year (R , if $R = 0$, evenly distributed; if $R = 1$, all floods occur on the same date) for river reaches of discharge value $\geq 1 \text{ m}^3 \text{ s}^{-1}$.

4. Despite the tradeoff, Grid LSTM-RAPID greatly outperforms a calibrated and bias-corrected process-based model, with a median KGE of 0.653 in the spatially and temporally out-of-sample test over small basins of good data quality and 0.592 globally over basins of all sizes and less data quality.
5. A new global reach-level daily database of discharge data set called GRADES-hydroDL (available at <https://www.reachhydro.org/home/records/grades-hydrodl>) is developed based on Grid LSTM-RAPID for the period from 1980 to near present (2025 so far) and offers great improvements against its predecessor.

The main point of this work is to test a modeling pathway that applies machine learning runoff estimation at grid cells and calculates river discharge over all river reaches with a routing model. Such a pathway works for its main goal (all-reach coverage) but with a price from extra uncertainties in HRU mismatch and in sub-basin scale processes like routing, for example:

1. We train LSTM with small basin per-area discharge and apply it to estimate runoff over finer (0.25°) grids. The grid areas fall within the range of areas of these training basins, supporting the feasibility of the application of the learned rainfall-runoff relationship. However, this HRU mismatch still introduces some errors, for example, the runoff response time at the finer grid cell could be shorter than at some small basins.
2. The rather simplified RAPID routing model, untuned against discharge observations, also introduces errors. It uses a fixed-velocity Muskingum method without explicit floodplain parameterization, and has known limitations in modeling flow dynamics in flat regions.
3. No human alterations to the river flows and water balance (dams, reservoirs, diversions, etc.) are modeled. The globe has a growing number of dams (Grill et al., 2019; Han et al., 2024; Zarfl et al., 2015), therefore the lack of dam and reservoir module may limit the simulation accuracy.

We believe these issues can be addressed and/or compensated with new emerging routing methods that can be better tuned against observed discharge and better adapted to human-altered river systems, for example, the differentiable routing techniques (Bindas et al., 2024). Moreover, future research could explore the potential of physics-informed ML paradigms (Wang et al., 2024), which combine the predictive capabilities of ML with physical consistency constraints, to enable global discharge estimation across all river reaches. Although this modeling pathway comes with challenges, the overall strategy is very promising.

Lastly, as much as GRADES-hydroDL is validated against available data, caution is highly advised in ungauged/less gauged areas, as well as in heavily managed basins. We encourage additional assessment of this product wherever possible.

Data Availability Statement

The MSWEP V2.80 precipitation data is available at MSWEP website (MSWEP, 2024). Other meteorological fields from ERA5 are available at Hersbach et al. (2017). The PROBA-V LAI is available at Copernicus Global Land Service website (CGLS, 2024). Elevation data from GMTED is available at Amatulli et al. (2018b). Geological permeability from GLHYMPS V2 is available at Huscroft et al. (2018b). Soil sand content data is from SoilGrids (Hengl et al., 2017). The underlying hydrography MERIT Basins for river routing is available at MERIT-Basins website (MERIT-Basins, 2024). Reservoir and dam data set is available at GRAND website (GRAND, 2024). Daily river discharge observations used in this study are compiled from diverse public sources. They are the Australian Bureau of Meteorology (BoM, 2024), CAMELS-BR (Chagas et al., 2020), CAMELS-CH (Höge et al., 2023), CAMELS-CL (Alvarez-Garretón et al., 2018), CAMELS-GB (Coxon et al., 2020), Czech Republic (CHMI, 2024), Denmark (ODA, 2024), Finland (2024), France (2024), the Global Runoff Data Centre (GRDC, 2024), Germany (BHS, 2024; SH, 2024), HYSETS (Arsenault et al., 2020), Ireland (EPA, 2024; OPW, 2024), Italy (ISPRA, 2024), Japan (MLIT, 2024), LamaH (Klingler et al., 2021), LamaH-Ice (Helgason & Nijssen, 2024), Poland (IMGW, 2024), Spain (2024), Sweden (SMHI, 2024), Thailand (RID, 2024), the United States Geological Survey (USGS) National Water Information System (NWIS, 2024). The training gauge list, global grid information, the evaluation gauges and their corresponding metrics are available in Zenodo repository (Yang & Pan, 2025). The final discharge data set generated in this study, namely GRADES-hydroDL (1980-near present) and its detailed description, is available at GRADES-hydroDL website (GRADES-hydroDL, 2024). The source codes for LSTM-based rainfall-runoff simulations are from hydroDL, which is available in Zenodo repository (Fang et al., 2021). The source codes for VIC model are available at VIC website (VIC, 2024). The

source codes for RAPID model are available in Zenodo repository (David, 2020). The evaluation codes used in this study are available in Zenodo repository (Yang & Pan, 2025).

Acknowledgments

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